

Cole Pearson

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EDUCATION

Shippensburg University, Shippensburg, PA

Candidate for Bachelor of Science in Chemistry (ACS Certified), Expected Graduation May 2026

Wood Honors College | Minor: Mathematics | GPA: 3.993 (4.0 without transfer credit)

RESEARCH & DISTINGUISHED ACADEMIC EXPERIENCES

Development of a Computational Titration Analysis Tool for Kombucha, May 2025–Present

PIs: Drs. Jeb Kegerreis, John Richardson, Department of Chemistry; Dr. Luis Melara, Department of Mathematics; Shippensburg University, Shippensburg, PA

- Developing an open-source computational tool to deconvolute weak acid mixtures
- Elucidating ionic effect evolution in complex organic acid mixtures
- Manuscripts in publication
- Awarded departmental *Beres Research Grant* (2025–2026) and *SURE Research Grant* (Summer 2025)

NSF REU: Sustainable Chemistry and Catalysis May 2024–July 2024; Continued Remote Collaboration

PI: Dr. Scott Sayres; Arizona State University, School of Molecular Sciences, Tempe, AZ

- Elucidated increasing excited-state lifetimes of bromoalkanes as a function of carbon chain length
- Operated Ti:Sapphire femtosecond laser, Nd:YAG 800 nm laser, time-of-flight mass spectrometer
- Employing *Gaussian* software to perform quantum mechanical energetic calculations
- Conducting data analysis and contributing to forthcoming peer-reviewed publication

DOE/ACS Nuclear Chemistry Summer School (NCSS), June 2025–July 2025

Lead Instructor: Jeff C. Bryan; San Jose State University, San Jose, CA

- **Trish Baisden Student of the Year Award**; graduated with A+ in both lecture and lab courses
- Completed intensive 6-week, 7-credit lab, lecture, and safety training program
- Studied fundamental nuclear science, fuel reprocessing, medicine, security, radiochemistry, and energy

MaterialZ Winter School, January 2026

Sponsored by ASU, & University of AZ; Biosphere 2, Oracle, AZ

- Materials Science & Engineering seminar series featuring industry and academic leaders
- Presented poster: “Development of a Computational Tool for the Analysis of Kombucha Acid–Base Titration Data”

German History, Culture, and Geography Course Development, May 2023–August 2023

Advisors: Dr. Alison Feeny & Dr. David Wildermuth, Shippensburg University, Shippensburg, PA

- Compiled course materials and assisted in curriculum development; Awarded SURE Grant (2023)
- Selected as one of top two presentations at Shippensburg University Summer Research Expo (2023)

AWARDS AND ACHIEVEMENTS

- **Dr. Louise J. Lienemann Chemistry Scholarship**, Shippensburg University, *Junior & Senior Year* "...awarded to a junior chemistry major demonstrating academic excellence... Character, citizenship, and a cumulative GPA of at least 3.5 on a 4.0 scale are factors in selecting the recipient."
- **Dr. James A. Beres and Merry Jean Beres Research Award**, Shippensburg University, 2025–2026 Competitive departmental grant awarded annually to one senior student for outstanding academic achievement, exceptional departmental service, and demonstrated potential in proposed research
- **Trish Baisden Student of the Year Award**, Nuclear Chemistry Summer School, 25 July 2025
- **ACS Standardized Exams (National Percentile Ranking)**
General Chemistry (97th) | Physical Chemistry (100th) | Organic Chemistry (96th) | Biochemistry (95th)

- **Shippensburg University Library Research Award, First Prize, 29 April 2023**
"The Use of Psychedelics in Therapies Targeting Alzheimer's Disease: A Literature Review"

PRESENTATIONS (* denotes regional or national conference/meeting)

- **Accepted: Minds@Work Symposium, 28 April 2026**
Shippensburg University, PA – "Development of a computational tool for kombucha titration analysis"
- **Accepted: Wood Honors College Senior Symposium, 21 April 2026**
Shippensburg University, PA – "Development of a computational tool for kombucha titration analysis"
- ***Accepted: Intercollegiate Student Chemists Convention Presentation Competition, 18 April 2026**
Ursinus College, Collegeville, PA – "Development of a computational tool for kombucha titration analysis"
- ***American Chemical Society (ACS) National Meeting Spring 2026, 22–26 March 2026**
Atlanta, GA – "Development of a computational tool for kombucha titration analysis"; Won competitive travel award (\$1k) for the **ACS–AGFD undergraduate poster competition**
- **Chemistry Club Guest Seminar, 2 February 2026**
Shippensburg University, PA – "Preparing Applications for REUs, Internships, and Grad School"
- ***National Collegiate Honors Council 2025, "Voices of Peace", 6–9 November 2025**
San Diego, CA – "Voices of Peace: Thinking Nobel and Acting Local in Shippensburg, PA", presented as co-founder of Shippensburg University Partners in Peace collaboration with Nobel Peace Center
- **Seminar, Department of Chemistry & Biochemistry, 10 October 2025**
Shippensburg University, PA – "NCSS in San José"
- ***Gulf Coast Undergraduate Research Symposium (GCURS) 2025, 18 October 2025**
Rice University, Houston, TX – "Development of a Computational Tool for Kombucha Titration Analysis"
- **Seminar, Department of Chemistry & Biochemistry, 8 November 2024**
Shippensburg University, PA – "REU at ASU: Ultrafast Laser Spectroscopy"
- **Summer Undergraduate Research Experience – Success Now Exposition, 22 August 2025**
Shippensburg University, PA – "German Beer History, Culture, and Geography Course Development"
- **NSF REU Oral Infographic Presentation, 21 July 2024**
Arizona State University, Tempe, AZ – Oral presentation and Q&A at School of Molecular Sciences
- **Minds@Work Symposium, 23 April 2024**
Shippensburg University, PA – "Chemistry and German Beer"
- ***Pennsylvania Undergraduate Research at the Capitol (URC), 19 March 2024**
State Capitol, Harrisburg, PA – "German Beer History, Culture, and Geography Course Development"
- **Summer Undergraduate Research Experience – Success Now Exposition, 25 August 2023**
Shippensburg University, PA – "German Beer History, Culture, and Geography Course Development"

SKILLS

- **Analytical techniques:** Femtosecond laser, gamma spectroscopy, alpha, IR, & NMR, UV/Vis spectroscopies; TOF mass spectrometry; gas & liquid chromatography; potentiometry
- **Class IV Laser Operation & Safety Certification:** Conducted alignment, diagnostics, maintenance
- **Radioactive Material Handling & Safety:** Completed 1-credit radiation safety course and 3-credit advanced nuclear chemistry lab; handled radioactive sources and conducted radionuclide syntheses
- **Biochemical skills:** SDS–PAGE, protein quantification assays, dialysis, ultracentrifugation, enzyme kinetics, and buffer preparation
- **General Laboratory Techniques:** Laser maintenance, organic synthesis, distillation, calorimetry, sample preparation, separations chemistry
- **Computational Skills:** MATLAB™ modeling and development; Python – data analysis, modeling, and workflow; *Gaussian* (software) – performed quantum mechanical density functional theory calculations
- **German language:** Intermediate proficiency; High school + 12 credits of college coursework

PROFESSIONAL EXPERIENCE

- **Upper Division Biochemistry I & II Teaching Assistant**, *August 2025–Present*, Shippensburg University
- **General & Organic Chemistry Tutor**, *August 2024–Present*, Shippensburg University
- **Honors Peer Anchor**, *August 2024–Present*, Shippensburg University
Leading first-year cohort through academic and professional transition to college; 10 hrs/week of mentorship
- **Package Handler**, *May 2022–March 2024*, UPS, Carlisle, PA
Worked 25–30 hours/week along full-time studies; first “Employee of the Month” among 500+ employees

CAMPUS INVOLVEMENT/COMMUNITY SERVICE

- **Founding Member, Partners in Peace, Nobel Peace Center Collaboration**, *January 2025–Present*
Launched “Voices of Peace” community outreach event; principal author of Nobel Peace Prize Nomination
- **Vice President, Chemistry Club**, *May 2025–Present*
Lead campus/community outreach; host fun events, alumni networking, and local school science nights
- **Treasurer, Kappa Mu Epsilon (Math Honor Society)**, *May 2025–Present*
- **Peer Mentor, Department of Chemistry & Wood Honors College**, *August 2024–Present*
- **Volunteer, ShipVotes** (nonpartisan voter participation group), *March 2024–Present*
- **Academic Appeals Committee Member, SU Department of Chemistry**, *August 2024–Present*
- **Volunteer Judge, Scholastic Art & Writing Awards**, *November 2024–Present*
- **Staff News Writer & DJ**, WSYC 88.7FM Radio, *September 2023–Present*
- **Volunteer, Honors Service Committee**, *August 2022–Present*
- **Chair, Orientation Committee**, Wood Honors College, *January 2023–April 2024*